

2 Smallpox, measles and HIV/AIDS are infectious diseases caused by different viruses. These different viruses share some structural features.

(a) State **one** structural feature that would confirm that a pathogen is a virus.

..... [1]

(b) The virus that causes smallpox belongs to a different genus to the virus that causes measles.

Name the viruses that cause these diseases.

smallpox .....

measles ..... [2]

(c) Explain why antibiotics, such as penicillin, **cannot** be used to treat measles.

.....  
.....  
.....  
.....  
..... [2]

(d) Antibiotics may be prescribed for a person with HIV/AIDS.

Suggest why antibiotics may be prescribed for a person with HIV/AIDs.

.....  
.....  
..... [1]

- A student correctly listed four reasons for the success of the global vaccination programme for smallpox. These reasons are listed in Fig. 2.1.

- Fig. 2.1**

- [4]

- Complete each row of Table 2.1 with a tick (✓) or a cross (X) to summarise the types of immunity gained by a person who received the smallpox vaccine.

| type of immunity    | gained (✓)<br>not gained (x) |
|---------------------|------------------------------|
| active immunity     |                              |
| artificial immunity |                              |
| natural immunity    |                              |
| passive immunity    |                              |

[Total: 11]